



SOLAR BRIGHT SYSTEM

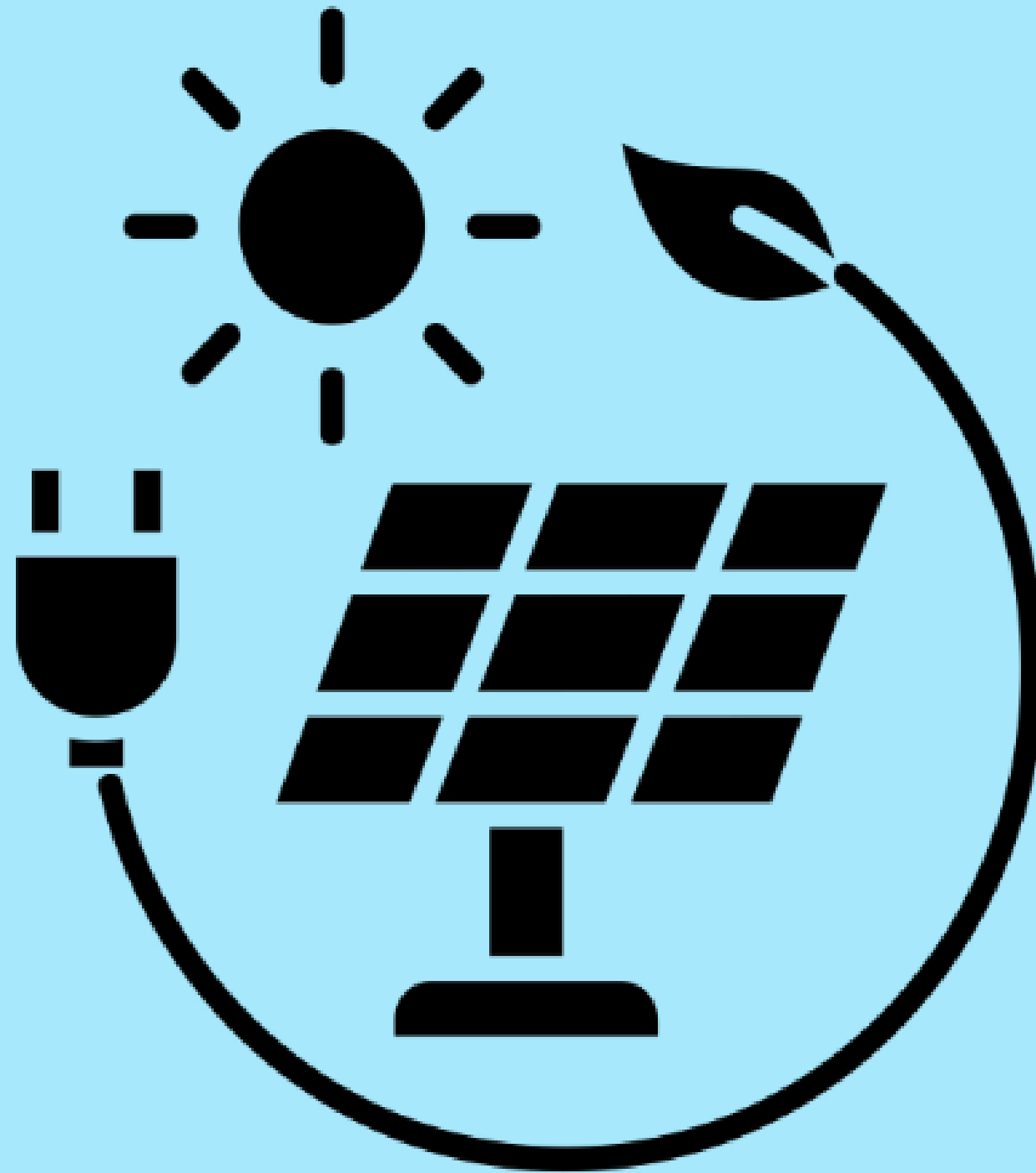
Track your sunshine , every hour





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INTRODUCTION

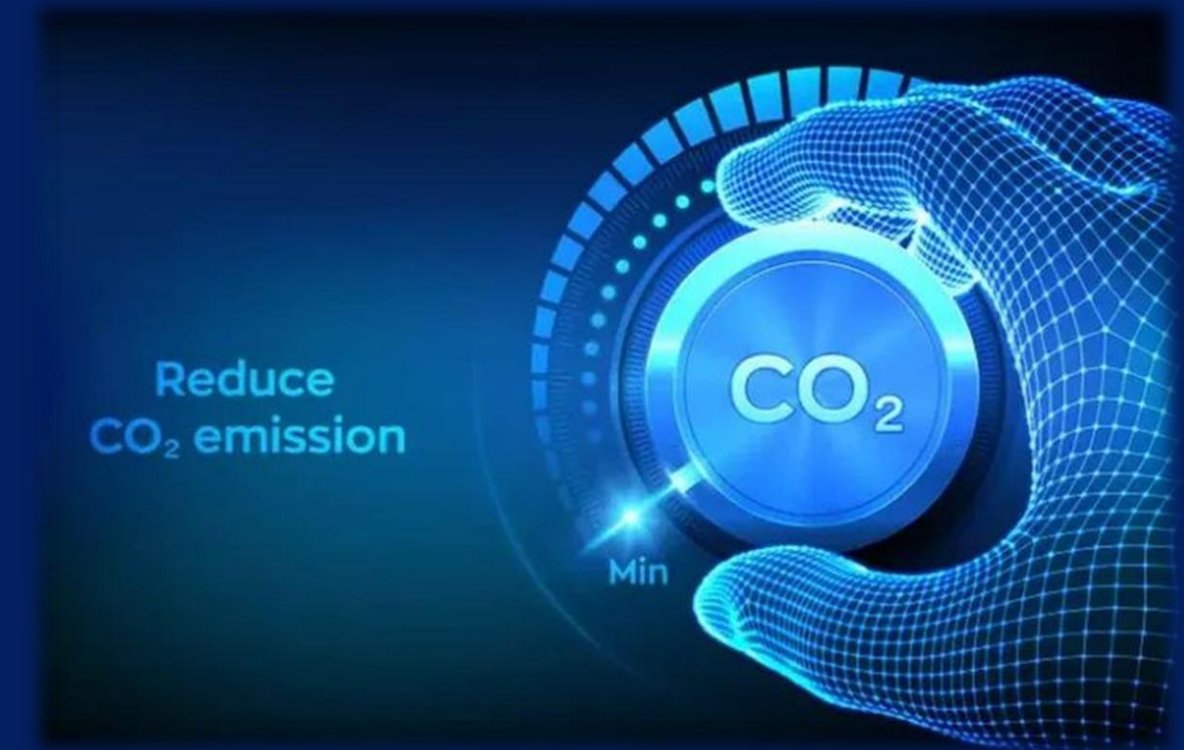
Electricity demand is continuously increasing, making efficient energy management important. Solar energy is a clean and sustainable source of electricity, but its generation can be affected by weather conditions. Users may not know in advance when unfavourable weather could reduce electricity generation. Therefore, our system combines solar monitoring, electricity consumption tracking, weather forecasting, billing, and solar services in one platform. The system is designed for government and citizens to improve electricity management and awareness.





ABSTRACT

This project presents a Solar Bright System. It monitors real-time solar power generation and city-wide electricity consumption. It uses weather forecasting to predict possible reductions in electricity supply and alert users in advance. Users can monitor their electricity usage, view and pay monthly electricity bills, and find government-approved solar companies. The system also displays the amount of carbon emissions reduced through renewable energy usage. Overall, the system aims to provide a smart, transparent, sustainable, and user-friendly electricity management platform.



8 DAY FORECAST							
MON	TUE	WED	THU	FRI	SAT	SUN	MON
60%	50%		30%				
25	26	28	27	28	30	32	25
15	16	18	22	22	21	15	

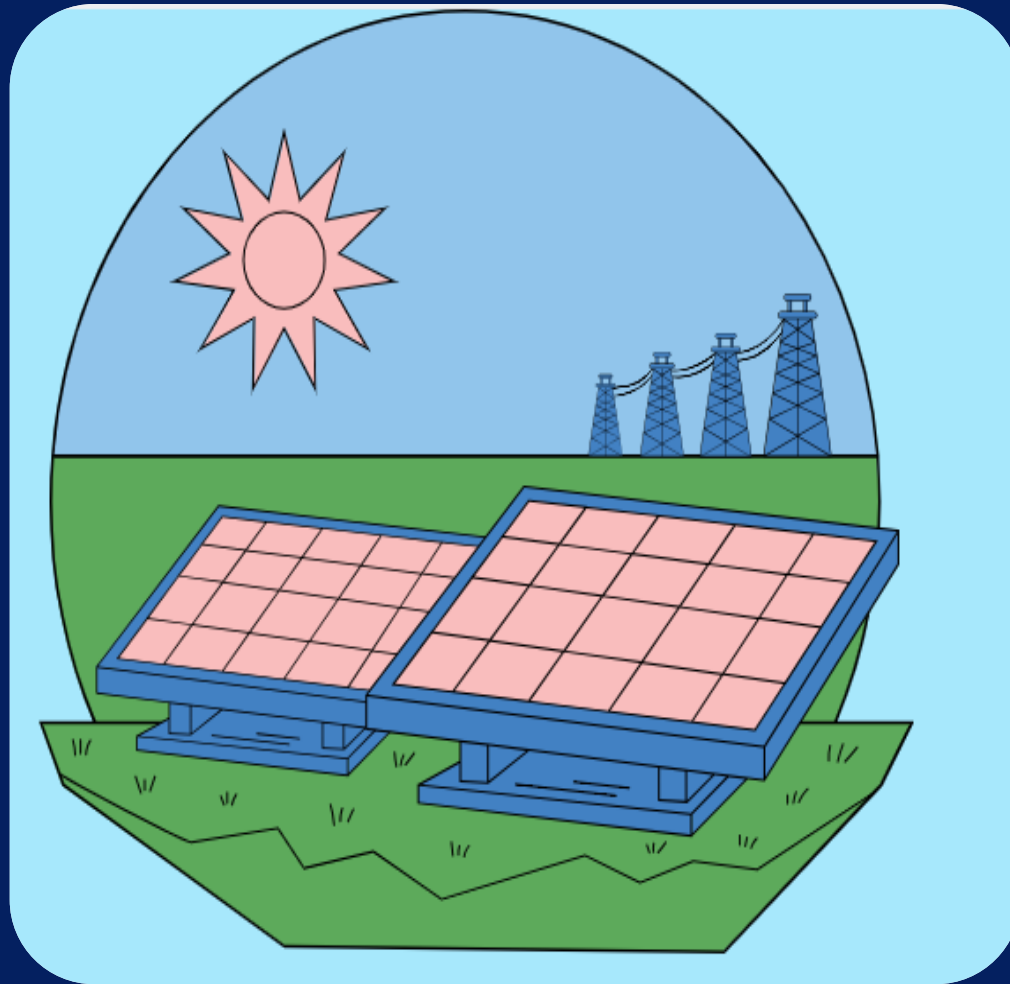


OBJECTIVES

- ⚡ To monitor current solar power generation (W/kW).
- 🌱 To show carbon emissions reduction from renewable energy usage.
- 📊 To track daily, weekly, and monthly solar generation.
- 🏠 To monitor city-wide electricity consumption.
- 📈 To compare energy generation and consumption.
- ☁️ To use weather forecasting to identify possible power reduction risks.
- 🔔 To provide early alerts to users about possible electricity supply reductions.
- 💰 To allow users to view and pay monthly electricity bills.
- ⚙️ To connect users with government-approved solar companies.



Advantages



- ⚡ Real-Time Monitoring – Track energy generation & consumption live (government & users)
- 🌱 Environmental Awareness – See carbon reduction data to encourage clean energy
- ☁️ Early Weather Alerts – Get advance warnings to prepare for power reductions
- 💳 Easy Bill Payment – View and pay monthly bills online in seconds
- ⚙️ Solar Accessibility – Find approved solar installation companies easily
- 🏠 All-in-One Centralized Platform – Access multiple electricity services in one web app
- 📊 Data Transparency – Clearly understand your usage and energy status



LIMITATIONS

- ⚠ Weather Alert Accuracy – Depends on the weather forecasting API and available data quality
- 📡 Real-Time Data Reliance – Requires reliable sensors or government electricity-system data
- 🌐 Internet Required – Needs stable connectivity to access the web app and receive updates
- 💳 Payment System Dependency – Online bill payment relies on supported third-party payment gateways
- 📋 Solar Company Updates – Must be regularly maintained by government or system admins
- 📊 Carbon Estimates Only – Reductions are approximate calculations based on usage & renewable data





CONCLUSION

The Solar Bright System provides an integrated platform for modern electricity management. It combines solar generation monitoring, electricity consumption tracking, weather forecasting, power alerts, bill payment, solar company services, and carbon reduction monitoring. By providing information and alerts in advance, the system can help users make better decisions about electricity usage. It also supports the government's goal of promoting efficient energy management and sustainable solar energy. In the future, the system can be expanded with AI-based energy prediction, smart-grid integration, and more advanced energy analytics.





THANK YOU